Regression Discontinuity Designs

Sharp and fuzzy designs, with a Stata exercise

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RDD seminar

Based on Scott Cunningham's *Causal Inference: The Remix* (second edition)

Seminar outline

1. Recognize a cutoff-based research design
2. Understand continuity and the sharp RD estimand
3. Estimate the discontinuity and assess credibility
4. Connect fuzzy RD to instrumental variables
5. Apply the method to DUI sanctions and recidivism in Stata

Presentation: 50–55 minutes **Exercise:** 20 minutes

Reading: Cunningham, Chapters 6 and 7 (fuzzy RDD subsection) (Cunningham, 2026)

A rule can create a research design

- A score X_i determines eligibility at a known threshold c.
- Compare the outcome *limits* immediately above and below c.
- The score can predict outcomes strongly; identification rests on what happens *at the cutoff*.

Assignment rule	Application in the manual
Test-score threshold	Merit awards (Thistlethwaite and Campbell, 1960)
Enrollment thresholds	Class size (Angrist and Lavy, 1999)
School boundaries	Housing prices (Black, 1999)

Cunningham, §6.1; overview: Lee and Lemieux (2010)

Legal access to alcohol at age 21

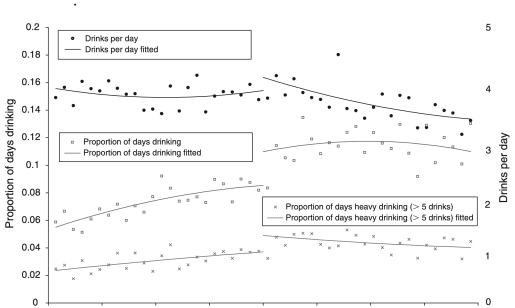


FIGURE 2. AGE PROFILE OF DRINKING INTENSITY

Notes: People can report their drinking for the last week, month, or year; 71 percent of respondents used a reference period of one week or one month. Average number of drinks per day is for people who reported some drinking.

Dots: age-bin averages. Curves: separate fits. The jump occurs at age 21.

Cunningham, Figure 6.2; original study: Carpenter and Dobkin (2009)

Mortality at the same threshold

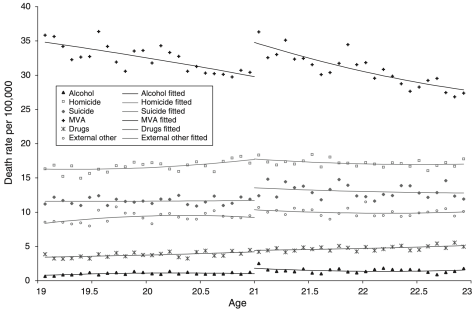


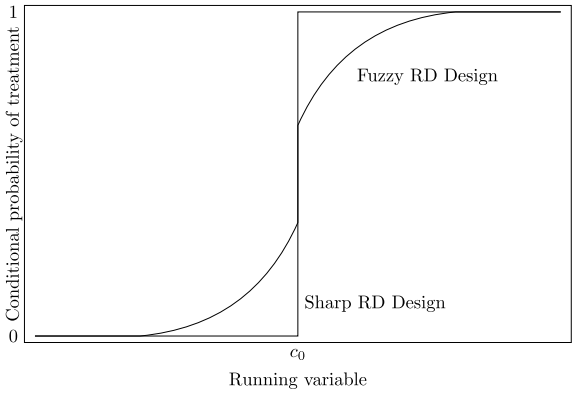
FIGURE 4. AGE PROFILES FOR DEATH RATES BY EXTERNAL CAUSE

Notes: See notes to Figure 3. The categories are mutally exclusive. The order of precedence is homicide, suicide, MVA, deaths with a mention of alcohol, and deaths with a mention of drugs. The ICD-9 and ICD-10 Codes are in Appendix C.

Mortality also jumps. The sharp treatment is **legal access**, not alcohol consumed.

Cunningham, Figure 6.3; (Carpenter and Dobkin, 2009)

Sharp and fuzzy assignment



Cunningham, Figure 6.4; adapted there from van der Klaauw (2002)

Sharp RD

$D_i = \mathbf{1}\{X_i \geq c\}$

Treatment changes from 0 to 1.

Fuzzy RD

$\Pr(D_i = 1 \mid X_i = x)$ jumps, with imperfect take-up.

Potential outcomes and observed outcomes

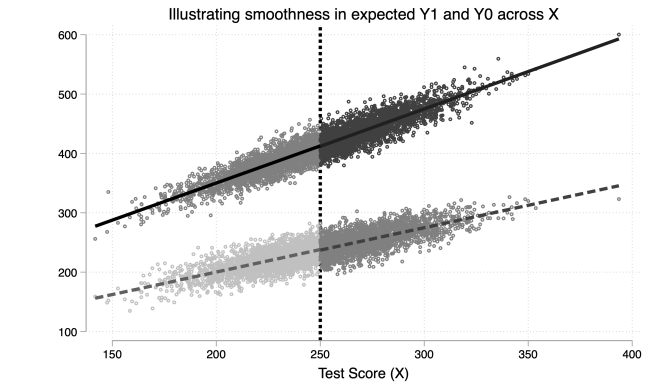
- Define treatment first: $D_i = 1$ means receiving the intervention.
- $Y_i(1)$ and $Y_i(0)$ exist for every unit, whatever its actual treatment.
- We observe only $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$.

Sharp assignment	Observed	Missing counterfactual
$X_i < c$	$Y_i(0)$	$Y_i(1)$
$X_i \geq c$	$Y_i(1)$	$Y_i(0)$

No treated and untreated units share the same score in a sharp design.

Cunningham, §6.2; supplied smoothness handout, pp. 1–3 (Cunningham, 2026)

Two potential-outcome curves



Cunningham, Figure 6.7; supplied smoothness handout (Hahn et al., 2001)

$$m_d(x) = \mathbb{E}[Y_i(d) \mid X_i = x]$$

Hold treatment fixed as you move horizontally.

Each curve describes group averages, not one person's trajectory.

Both schedules are visible here because the data are simulated.

The continuity assumption

For each fixed treatment status $d \in \{0, 1\}$:

$$\lim_{x \to c^-} \mathbb{E}[Y_i(d) \mid X_i = x] = \lim_{x \to c^+} \mathbb{E}[Y_i(d) \mid X_i = x] = m_d(c).$$

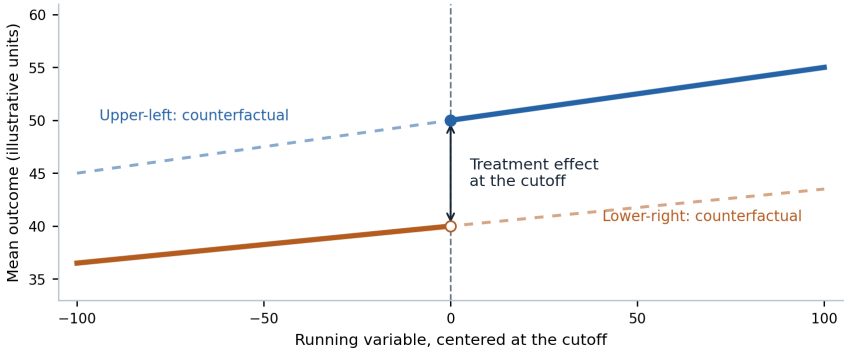
Read horizontally

Applicants on opposite sides approach the same mean outcome under the *same hypothetical treatment status*.

- Each potential-outcome mean is continuous at c.
- Curves may slope, bend, and have different slopes.
- Continuity neither implies a zero treatment effect nor equal individual outcomes.

Cunningham, §6.2; supplied handout, p. 2 (Hahn et al., 2001)

Why the observed outcome can jump



- The observed mean follows $m_0(x)$ below c and $m_1(x)$ above c.
- The vertical gap at c is $m_1(c) - m_0(c)$.

Supplied smoothness handout, p. 3; schematic values are illustrative (Hahn et al., 2001)

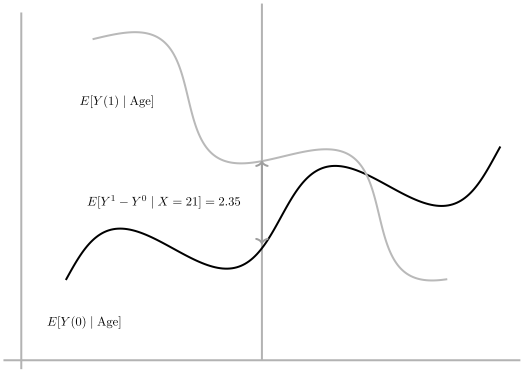
Identification at the cutoff

$$\begin{aligned}\tau_{\text{SRD}}(c) &= \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = c] \\ &= \underbrace{\lim_{x \to c^+} \mathbb{E}[Y_i \mid X_i = x]}_{m_1(c)} - \underbrace{\lim_{x \to c^-} \mathbb{E}[Y_i \mid X_i = x]}_{m_0(c)}.\end{aligned}$$

- Assignment tells us which outcome we observe on each side.
- Continuity connects those observed limits to the missing means at c.
- This identifies the **ATE at the cutoff**.

Cunningham, §6.2; (Hahn et al., 2001; Imbens and Lemieux, 2008)

A local effect need not generalize



Continuous curves can imply different effects at different scores. RD identifies the gap at c.

Cunningham, Figure 6.6 (Cunningham, 2026; Lee and Lemieux, 2010)

A university example for the counterfactuals

Teaching simplification: everyone above the admission cutoff attends the flagship.

$$X_i = \text{SAT}_i - \text{required SAT}(\text{GPA}_i, \text{cohort}_i), \qquad c = 0.$$

- $Y_i(1)$: earnings if applicant i attends the flagship.
- $Y_i(0)$: earnings if the same applicant does not attend it; another university is possible.
- Imagine *nobody* attended: mean earnings should pass continuously through zero.
- Imagine *everybody* attended: that mean should also pass continuously through zero.

Supplied handout, pp. 4–5. The actual study is fuzzy RD (Hoekstra, 2009)

Continuity with a positive treatment effect

Adjusted SAT	Without flagship	With flagship
−10	\$39,650 (observed)	\$49,500 (missing)
0	\$40,000 (missing)	\$50,000 (observed)
+10	\$40,350 (missing)	\$50,500 (observed)

Both schedules are continuous. The cutoff effect is **\$10,000**. The observed difference between −10 and +10 is \$10,850: it also reflects differences associated with score.

Invented dollar amounts from the supplied handout. Hoekstra analyzes log earnings; these are not paper estimates (Hoekstra, 2009)

OLS with separate slopes

Center the running variable: $x_i = X_i - c$.

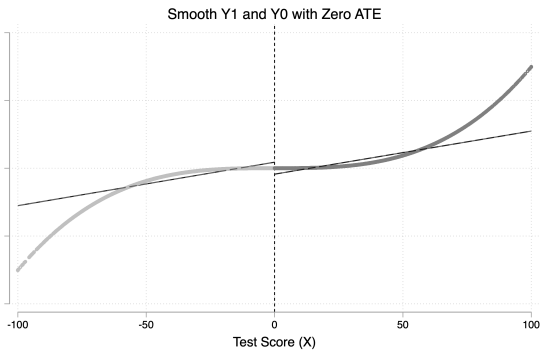
$$Y_i = \alpha + \tau D_i + \beta x_i + \gamma D_i x_i + \varepsilon_i.$$

- The left intercept is α; the right intercept is $\alpha + \tau$.
- The slopes are β and $\beta + \gamma$.
- $\hat{\tau}$ is the fitted gap *at the cutoff*.

```
generate double x_c = score - cutoff
generate byte treated = (score >= cutoff) if !missing(score)
regress outcome i.treated##c.x_c if abs(x_c) <= h, vce(robust)
```

Generic Stata syntax; h is a chosen window. Cunningham, §6.3 (Cunningham, 2026)

Functional form can create a false jump



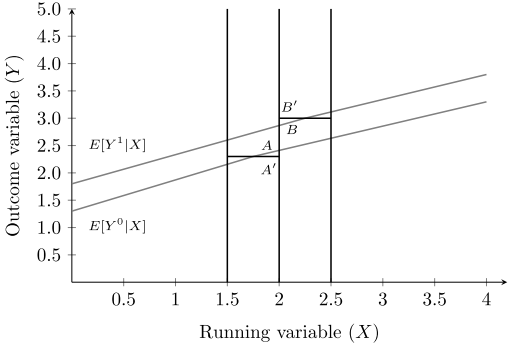
Cunningham, Figure 6.9 and Table 6.1 (Gelman and Imbens, 2019)

The manual simulates **zero treatment effect** with a nonlinear outcome-score relationship.

Broad-window straight lines give misleading cutoff predictions.

Adding high global polynomial orders is unstable; use local low-order fits.

Bin averages and boundary bias



A bin average summarizes outcomes away from the cutoff. With trends, its difference need not equal the boundary gap. **Binning is for display; estimation uses individual observations.**

Cunningham, Figure 6.10; binscatter discussion: Cattaneo et al. (2024)

Local linear regression

Fit a weighted regression separately on each side:

$$(\hat{a}_s, \hat{b}_s) = \arg\min_{a,b} \sum_{i \in s,\, |x_i| \le h_s} K(x_i/h_s)(Y_i - a - bx_i)^2.$$

$$\hat{\tau} = \hat{a}_+ - \hat{a}_-, \qquad K(u) = (1 - |u|)\mathbf{1}\{|u| \le 1\}.$$

- **Local:** use a neighborhood around c.
- **Linear:** allow different slopes on either side.
- **Triangular weights:** give nearby observations more influence.

Cunningham, §6.3 (Hahn et al., 2001; Calonico et al., 2014; Cattaneo et al., 2020a)

Bandwidth, kernel, and polynomial order

- **Bandwidth h:** a narrower window usually reduces approximation bias but uses less information.
- **Kernel K:** determines observation weights inside the window.
- **Order p:** local linear ($p = 1$) is a standard starting point.

$$\text{MSE}(\hat{\tau}) = \text{Bias}(\hat{\tau})^2 + \text{Var}(\hat{\tau}).$$

Data-driven bandwidths balance bias and variance; they can differ by side. Report the chosen windows and examine sensitivity.

Cunningham, §6.3 (Imbens and Kalyanaraman, 2012; Calonico et al., 2014)

Robust bias-corrected inference

At an MSE-optimal bandwidth, approximation bias can matter for confidence intervals.

Interval	Center	Standard error
Conventional	$\hat{\tau}$	Conventional
Bias-corrected	$\hat{\tau} - \hat{B}$	Conventional
Robust bias-corrected	$\hat{\tau} - \hat{B}$	Includes bias estimation

$$\mathrm{CI}_{\mathrm{RBC}} = (\hat{\tau} - \hat{B}) \pm 1.96\,\mathrm{SE}_{\mathrm{RBC}}.$$

In `rdrobust`, pair `e(tau_bc)` with `e(se_tau_rb)`.

Cunningham, §6.3 (Calonico et al., 2014; Calonico et al., 2017)

What can invalidate the comparison?

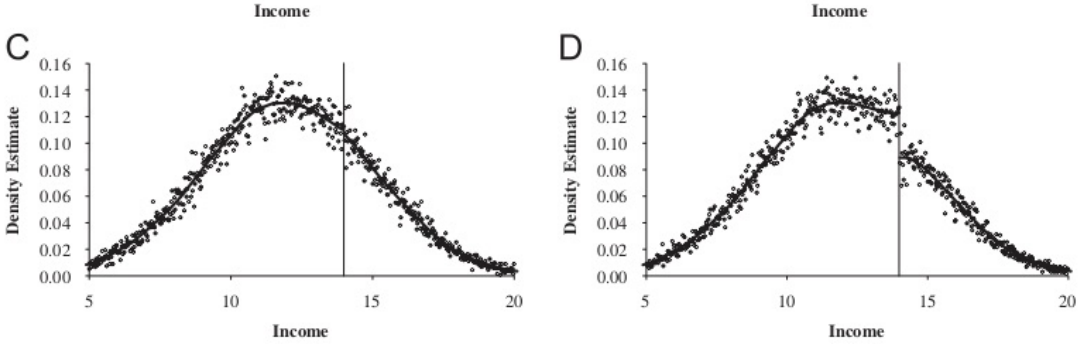
- Precise sorting: people manipulate their scores around c.
- Another intervention changes at exactly the same cutoff.
- Measurement, sample selection, or reporting rules change there.
- Nonrandom heaping makes neighboring scores different types of units.

A failure of continuity

If untreated potential outcomes jump at c, the observed discontinuity can combine that jump with the treatment effect.

Cunningham, §6.4 (Imbens and Lemieux, 2008; Lee and Lemieux, 2010)

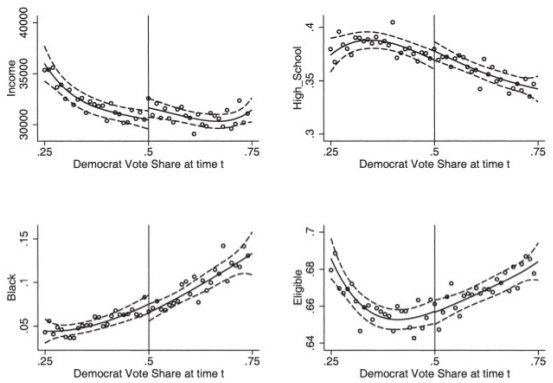
Density continuity and manipulation



- Look for a change in the distribution of scores at c.
- A density test is evidence about sorting; it cannot verify missing potential outcomes.

Cunningham, Figure 6.11 (McCrary, 2008; Cattaneo et al., 2020b)

Predetermined covariates and placebo cutoffs



In close elections, check whether district characteristics jump at 50% Democratic vote share.

Also estimate jumps at placebo cutoffs within each side.

Null findings support credibility; they do not establish continuity.

Cunningham, Figure 6.22 (covariate checks) (Lee et al., 2004; Imbens and Lemieux, 2008)

Birth-weight heaping

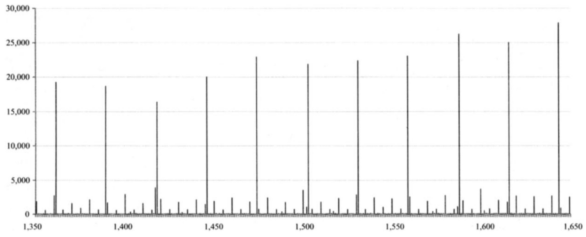


FIGURE I
Frequency of Births by Gram: Population of U.S. Births between 1,350 and 1,650 g
NCHS birth cohort linked birth/infant death files, 1983–1991 and 1995–2003, as described in the text.

Very-low-birth-weight classification changes at 1,500 grams.

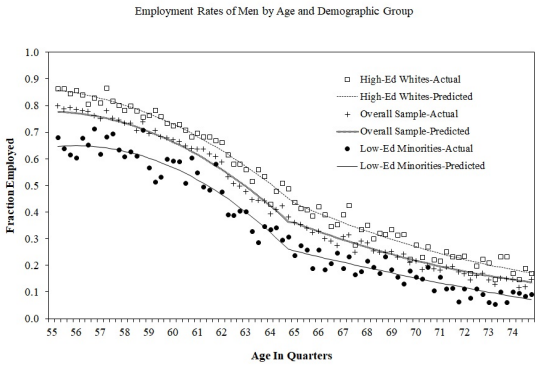
Reported weights heap at round values.

If heaped babies differ systematically, the local comparison can be biased.

Excluding the cutoff changes the sample and needs justification.

Cunningham, Figure 6.13 (Almond et al., 2010; Barreca et al., 2011; Barreca et al., 2016)

Medicare eligibility at age 65



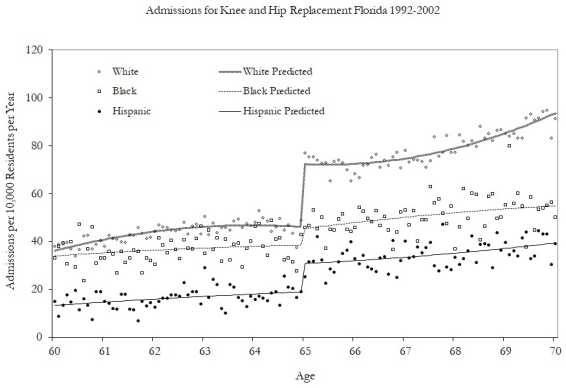
Cunningham, §6.5 and Figure 6.14 (Card et al., 2008)

Age 65 changes eligibility sharply; actual insurance coverage has imperfect take-up.

Retirement is a potential competing mechanism.

The manual uses employment profiles to assess whether work changes abruptly at 65.

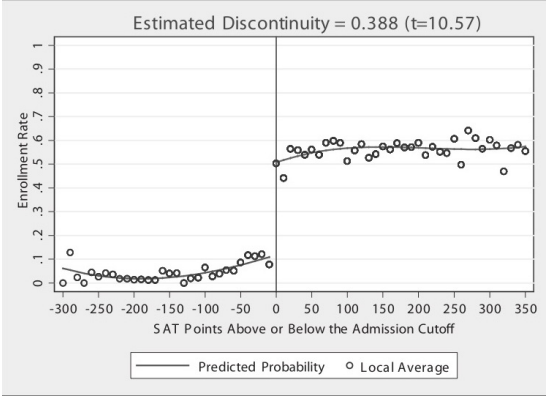
Healthcare utilization at age 65



This is a reduced-form effect of crossing the eligibility threshold. An effect of actual coverage requires a first stage and additional assumptions.

Cunningham, Figure 6.15; utilization and subsequent mortality study (Card et al., 2008; Card et al., 2009)

Hoekstra's actual design: admission and attendance



Cunningham, Figure 6.20; supplied handout, pp. 4–5 (Hoekstra, 2009)

$Z_i = \mathbf{1}\{X_i \geq 0\}$: crossing the admission threshold.

D_i: attending the flagship.

Admission causes a jump in enrollment, rather than universal attendance.

Unpublished, changing cutoffs make precise targeting less plausible.

The fuzzy RD ratio

$$\tau_{\text{FRD}}(c) = \frac{\lim_{x \to c^+} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \to c^-} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \to c^+} \mathbb{E}[D_i \mid X_i = x] - \lim_{x \to c^-} \mathbb{E}[D_i \mid X_i = x]}.$$

- **Numerator:** reduced-form jump in the outcome.
- **Denominator:** first-stage jump in treatment take-up.
- **Illustration:** an outcome jump of 2 and a take-up jump of 0.4 imply a ratio of 5.

Illustrative numbers. Cunningham, §7.7, fuzzy RDD subsection (Hahn et al., 2001; Imbens and Lemieux, 2008)

Additional assumptions and the estimand

- **Continuity:** the relevant potential outcomes and compliance types vary smoothly at c.
- **Relevance:** crossing the cutoff changes the probability of treatment.
- **Exclusion:** crossing the cutoff affects Y through D.
- **Monotonicity:** crossing to the eligible side does not reduce anyone's take-up.
- **Well-defined treatment and no interference.**

The ratio identifies the **LATE for compliers at the cutoff**.

Cunningham, fuzzy RDD subsection (Hahn et al., 2001; Imbens and Angrist, 1994; Angrist et al., 1996)

Fuzzy RD as local 2SLS

With $x_i = X_i - c$ and $Z_i = \mathbf{1}\{X_i \geq c\}$:

$$D_i = a + \pi Z_i + bx_i + gZ_i x_i + v_i,$$
$$Y_i = \alpha + \tau D_i + \beta x_i + \gamma Z_i x_i + u_i.$$

Instrument D_i with Z_i; retain the same score controls on each side.

```
generate double zx = z*x_c
ivregress 2sls earnings x_c zx (attend = z) ///
    if abs(x_c) <= h, vce(robust)
* Alternatively: local polynomial fuzzy RD with RBC inference
rdrobust earnings score, c(0) fuzzy(attend) p(1)
```

Generic syntax, not a run on Hoekstra data. Cunningham, fuzzy RDD subsection (Calonico et al., 2017)

Another fuzzy design: Maimonides' rule

Angrist and Lavy: classes in Israeli schools are split at enrollment thresholds. (Angrist and Lavy, 1999)

- Rule-predicted class size falls when enrollment crosses a threshold.
- Actual class size responds imperfectly, creating a first stage.
- Outcomes are student achievement measures.
- The IV logic asks whether the rule changes achievement through class size.

Class size is a *continuous* treatment: the binary-treatment complier interpretation requires adaptation.

Cunningham, §6.1–6.2 and fuzzy RDD subsection (Cunningham, 2026)

Sharp and fuzzy RD: what is identified?

	Sharp RD	Fuzzy RD
Assignment	Treatment is determined by crossing c	Treatment probability changes at c
Estimator	Outcome discontinuity	Outcome jump / take-up jump
Estimand	ATE at c	Complier LATE at c
Main requirements	Continuity, well-defined treatment	Continuity plus IV assumptions

For either design, define treatment precisely and inspect the institutional rule.

Cunningham, Chapters 6–7 (Hahn et al., 2001; Lee and Lemieux, 2010)

Exercise: DUI sanctions and recidivism

Research question: does crossing the DUI BAC threshold reduce subsequent offending? (Hansen, 2015)

- X_i: measured blood alcohol content (BAC); cutoff $c = 0.08$.
- $D_i = \mathbf{1}\{X_i \geq 0.08\}$: exposure to the threshold-based DUI regime.
- Y_i: recidivism indicator in the teaching extract.
- Cunningham's extract has **214,558 observations**; this is a teaching exercise rather than a full replication of the paper.

20-minute task: assess credibility, estimate the gap, and interpret its magnitude.

Cunningham, §6.6; sanctions and deterrence framework (Cunningham, 2026; Becker, 1968)

Load the data and center BAC

Run directly in Stata 18. If necessary, select this do-file in the folder-location dialog:

```
do "hansen_seminar.do"
```

The script downloads once and then uses the cached data:

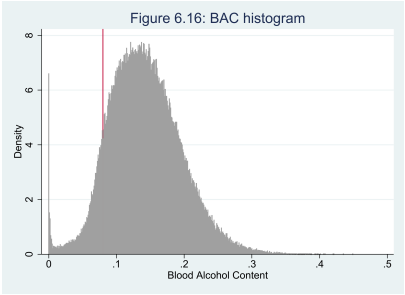
```
use "`output_dir'/hansen_dwi.dta", clear
rename bac1 bac1_orig
generate double bac1 = bac1_orig - 0.08
generate byte dui = (bac1_orig >= 0.08) if !missing(bac1_orig)
generate double bac1_sq = bac1^2
```

- Use `bac1_orig` with `c(0.08)`; centered BAC has cutoff zero.
- The script sets its folder with `cd`, then saves figures, tables, and a log.

Seminar script adapted from the supplied `hansen_complete.do` (Hansen, 2015; Cunningham, 2026)

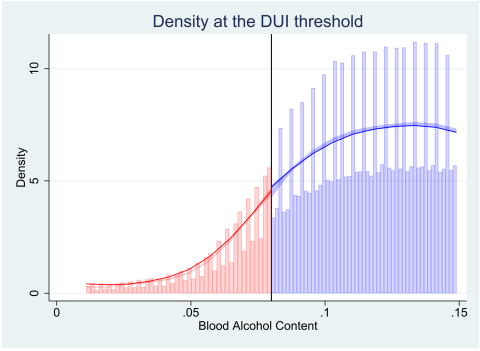
Inspect the BAC distribution

```
histogram bac1_orig, discrete width(0.001) color(gs10%90) ///
    xline(0.08) title("Figure 6.16: BAC histogram")
```



Generated from the teaching extract; compare Cunningham, Figure 6.16 (Hansen, 2015)

The density test and recorded precision



Robust density-test $p = 0.8897$.

It does not reject density continuity.

Narrow-window binomial tests reject at several widths; recorded precision matters.

```
rddensity bac1_orig, c(0.08) plot
```

Generated output. Interpret the tests with the histogram (McCrary, 2008; Cattaneo et al., 2020b)

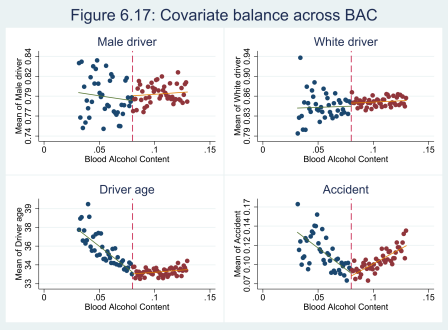
Covariate balance: code and estimates

```
foreach outcome in white male aged acc {
    regress ‘outcome’ i.dui##c.bac1 ///
        if inrange(bac1_orig, 0.03, 0.13), vce(robust)
}
```

	White	Male	Age	Accident
Discontinuity	0.0057	0.0062	-0.1405	-0.0034
Robust SE	0.0050	0.0057	0.1644	0.0041
Sample mean	0.8526	0.7919	34.1701	0.1012
Untreated cutoff level	0.8463	0.7842	33.9199	0.0834
Observations	89,967	89,967	89,967	89,967

Generated Table 6.6; proportions for White/Male/Accident, years for Age (Hansen, 2015)

Covariate balance: figures

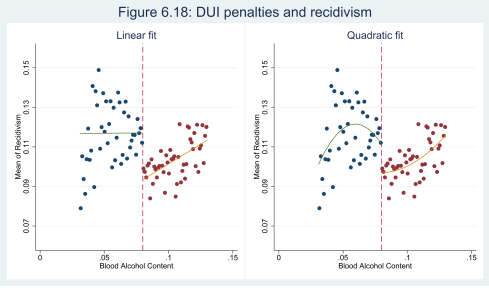



```
cmogram male bac1_orig if inrange(bac1_orig, 0.03, 0.13), ///
    scatter cut(0.08) line(0.08) lfit
```

Repeat for White, Age, and Accident; generated Figure 6.17 (Hansen, 2015)

Plot the recidivism discontinuity

```
cmogram recidivism bac1_orig if inrange(bac1_orig, 0.03, 0.13), ///
    scatter cut(0.08) line(0.08) lfit
* Repeat with qfit for the quadratic panel.
```



Generated Figure 6.18; read the fitted gap at 0.08 (Hansen, 2015)

OLS results in the fixed BAC window

```
regress recidivism i.dui##c.bac1 ///
    if inrange(bac1_orig, 0.03, 0.13), vce(robust)
regress recidivism i.dui##c.(bac1 bac1_sq) ///
    if inrange(bac1_orig, 0.03, 0.13), vce(robust)
```

	Linear	Quadratic
Effect (pp)	-2.362	-1.432
Robust SE (pp)	0.436	0.623
Sample recidivism (%)	10.694	10.694
Untreated cutoff level (%)	11.707	11.134
Observations	89,967	89,967

Generated Table 6.7: sample mean and fitted untreated cutoff level are distinct (Hansen, 2015)

Local polynomial RD: code

```
rdrobust recidivism bac1_orig, c(0.08) p(1) ///
    kernel(triangular) bwselect(msetwo) ///
    vce(nn 3) masspoints(adjust) all
```

- Repeat for $p = 0, 1, 2$: local constant, linear, and quadratic.
- Use the full BAC sample for bandwidth selection.
- `msetwo` chooses separate MSE-optimal windows by side.
- Pair the bias-corrected effect with the RBC standard error.

The script stores each model and its bandwidths (Calonico et al., 2014; Calonico et al., 2017)

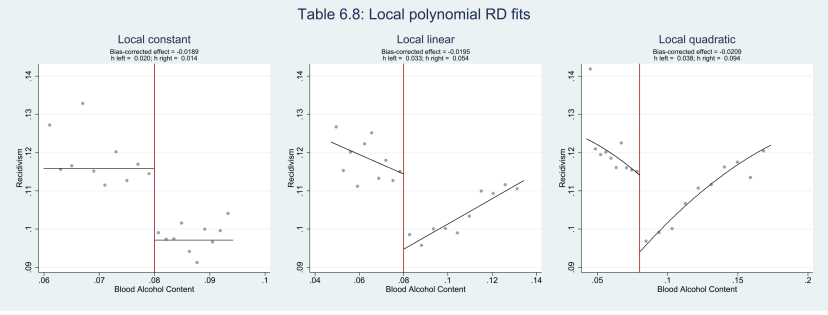
Local polynomial RD: results

	Constant	Linear	Quadratic
Bias-corrected effect (pp)	-1.891	-1.955	-2.093
RBC SE (pp)	0.507	0.607	0.712
Polynomial order	0	1	2
Bandwidth left	0.0201	0.0329	0.0378
Bandwidth right	0.0143	0.0544	0.0938
Effective N left	13,794	16,774	17,545
Effective N right	16,352	77,539	135,457
Effective N total	30,146	94,313	153,002

The quadratic right window reaches BAC 0.174, beyond the aggravated-DUI threshold of 0.15. Check sensitivity to narrower windows.

Generated Table 6.8: both sides and their sum are reported (Hansen, 2015; Calonico et al., 2017)

The local polynomial fits



- Each panel uses its own estimated left and right bandwidths.
- Curves show conventional fits; subtitles show bias-corrected effects.
- Binned dots summarize the data; they do not determine the estimates.

Generated from the stored Table 6.8 models (Hansen, 2015; Calonico et al., 2017)

Interpret the local linear result

Local linear estimate

A reduction of **1.95 percentage points** in recidivism.

Robust bias-corrected 95% CI: $[-3.14, \ -0.77]$ percentage points.

- This is a local effect of crossing the threshold-based DUI regime.
- It combines the sanctions and other consequences triggered there.
- The data do not separately identify each mechanism behind reduced recidivism.

Generated $p = 1$ result from Cunningham's teaching extract (Hansen, 2015)

Exercise discussion

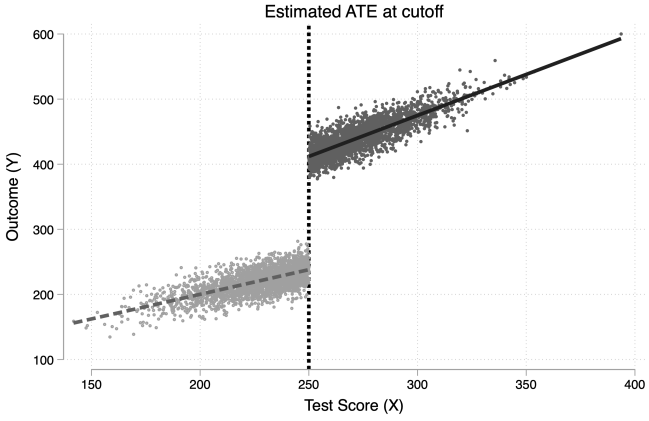
1. What makes continuity plausible in this institutional setting?
2. What do the histogram, density test, and balance checks establish?
3. Why do OLS and local linear estimates differ?
4. Which people does the estimate describe, and what does it say about mechanism?

Evidence to take away

A known rule, a visible outcome gap, credible local comparisons, and inference that accounts for approximation bias.

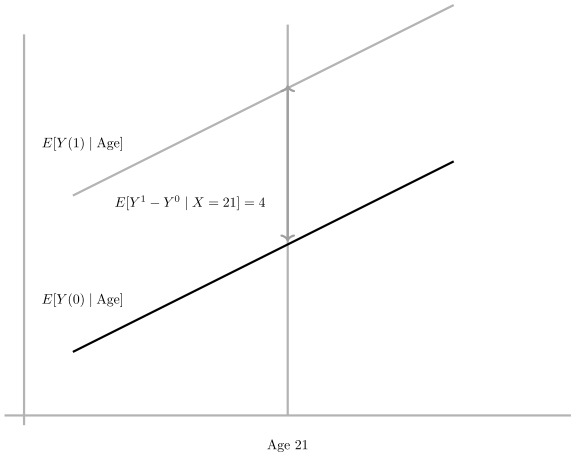
Cunningham, §6.6–6.7 (Imbens and Lemieux, 2008; Lee and Lemieux, 2010)

Observed data from the manual's simulation



Cunningham, Figure 6.8: observing different potential outcomes on either side (Cunningham, 2026)

Homogeneous effects as a special case



Cunningham, Figure 6.5; broader representativeness requires further assumptions (Cunningham, 2026)

Discrete scores and standard errors

- Many observations can share a score without providing many distinct score values near c.
- With fixed support and gaps, a continuous-score limiting argument needs additional structure.
- `masspoints(adjust)` helps bandwidth selection; it does not resolve all identification and inference issues from discreteness.
- Clustering only by running-variable values is not a general solution.

Cunningham, §6.3; early and later treatments (Lee and Card, 2008; Kolesar and Rothe, 2018)

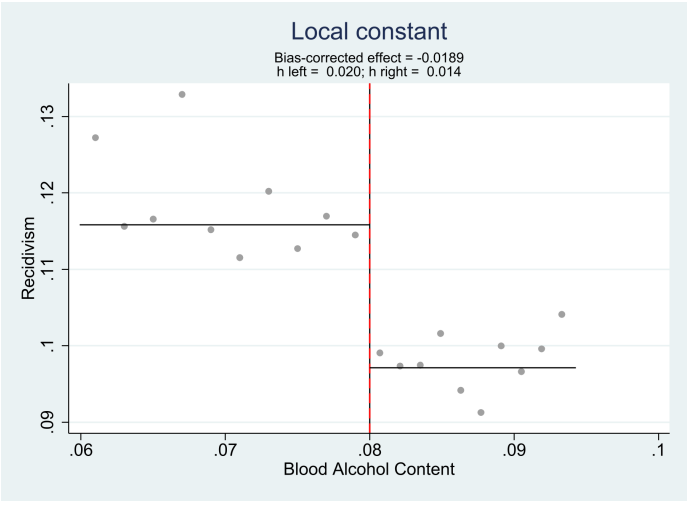
Plot the model's actual estimation window

```
estimates restore recid_rd_p1
local hl = e(h_l)
local hr = e(h_r)
local xmin = e(c) - `hl'
local xmax = e(c) + `hr'
rdplot recidivism bac1_orig if inrange(bac1_orig, `xmin', `xmax'), ///
    c(0.08) p(1) h(`hl' `hr') kernel(triangular) ///
    nbins(10 10) masspoints(adjust)
```

This reuses the stored model's bandwidths and the same polynomial order.

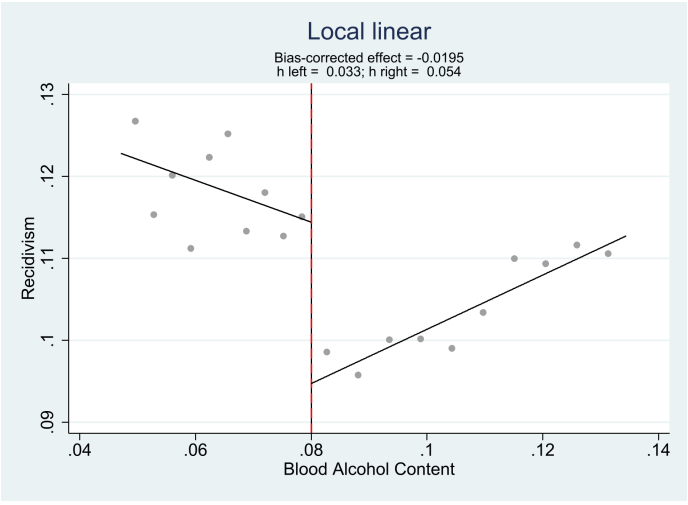
Seminar script, section 7 (Calonico et al., 2017)

Local constant fit



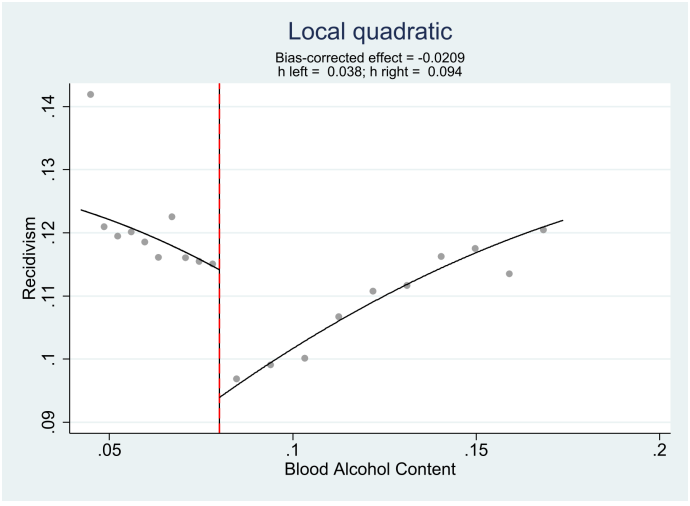
Generated from the Table 6.8 model with $p = 0$ (Hansen, 2015)

Local linear fit



Generated from the Table 6.8 model with $p = 1$ (Hansen, 2015)

Local quadratic fit



Generated from the Table 6.8 model with $p = 2$ (Hansen, 2015)

Store the estimated effects and uncertainty

```
forvalues p = 0/2 {
    local col = `p' + 1
    capture program drop rdrobust
    rdrobust recidivism bac1_orig, c(0.08) p(`p') ///
        kernel(triangular) bwselect(msetwo) ///
        vce(nn 3) masspoints(adjust) all
    estimates store recid_rd_p`p'
    matrix table6_8[1, `col'] = e(tau_bc)
    matrix table6_8[2, `col'] = e(se_tau_rb)
}
```

Excerpt from the runnable seminar script; its matrix also stores bandwidths and effective sample sizes (Calonico et al., 2017)

Export the figures and tables

```
graph save fig616 "‘output_dir’/figure_6_16.gph", replace
graph export "‘output_dir’/figure_6_16.png", ///
    name(fig616) width(2400) replace
* Example for one table:
preserve
clear
svmat double table6_7, names(col)
export delimited using "‘output_dir’/table_6_7_values.csv", replace
restore
```

The full script additionally exports descriptive row labels for all three tables.

Figures remain editable in Stata's GPH format. The deck's table inputs are generated from CSV output.

Other deterrence studies in the manual

- An Italian collective pardon changes the sentence faced after reoffending. (Drago et al., 2009)
- The adult criminal-justice threshold at age 18 offers another RD setting. (Lee and McCrary, 2017)
- These designs distinguish behavioral responses to threatened sanctions from the mechanical effects of incarceration.

Cunningham, §6.6; supplementary context for the Hansen exercise (Becker, 1968)

References I

Almond, D., J. Doyle Joseph J., A. E. Kowalski, and H. Williams (2010). "Estimating Marginal Returns to Medical Care: Evidence from At-Risk Newborns". In: *Quarterly Journal of Economics* 125.2, pp. 591–634.

Angrist, J. D., G. W. Imbens, and D. B. Rubin (1996). "Identification of Causal Effects Using Instrumental Variables". In: *Journal of the American Statistical Association* 91.434, pp. 444–455.

Angrist, J. D. and V. Lavy (1999). "Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement". In: *Quarterly Journal of Economics* 114.2, pp. 533–575.

Barreca, A. I., M. Guldi, J. M. Lindo, and G. R. Waddell (2011). "Saving Babies? Revisiting the Effect of Very Low Birth Weight Classification". In: *Quarterly Journal of Economics* 126.4, pp. 2117–2123.

Barreca, A. I., J. M. Lindo, and G. R. Waddell (2016). "Heaping-Induced Bias in Regression-Discontinuity Designs". In: *Economic Inquiry* 54.1, pp. 268–293.

Becker, G. S. (1968). "Crime and Punishment: An Economic Approach". In: *Journal of Political Economy* 76.2, pp. 169–217.

Black, S. E. (1999). "Do Better Schools Matter? Parental Valuation of Elementary Education". In: *Quarterly Journal of Economics* 114.2, pp. 577–599.

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik (2017). "rdrobust: Software for Regression-Discontinuity Designs". In: *Stata Journal* 17.2, pp. 372–404.

Calonico, S., M. D. Cattaneo, and R. Titiunik (2014). "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs". In: *Econometrica* 82.6, pp. 2295–2326.

Card, D., C. Dobkin, and N. Maestas (2008). "The Impact of Nearly Universal Insurance Coverage on Health Care Utilization: Evidence from Medicare". In: *American Economic Review* 98.5, pp. 2242–2258.

References II

Card, D., C. Dobkin, and N. Maestas (2009). "Does Medicare Save Lives?" In: *Quarterly Journal of Economics* 124.2, pp. 597–636.

Carpenter, C. and C. Dobkin (2009). "The Effect of Alcohol Consumption on Mortality: Regression Discontinuity Evidence from the Minimum Drinking Age". In: *American Economic Journal: Applied Economics* 1.1, pp. 164–182.

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng (2024). "On Binscatter". In: *American Economic Review* 114.5, pp. 1488–1514.

Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020a). *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press.

Cattaneo, M. D., M. Jansson, and X. Ma (2020b). "Simple Local Polynomial Density Estimators". In: *Journal of the American Statistical Association* 115.531. Published online in 2019; cited as 2019 in the manual, pp. 1449–1455.

Cunningham, S. (2026). *Causal Inference: The Remix. Online second edition*. Reading copy supplied for this seminar; Chapters 6 and 7. URL: `https://mixtape.scunning.com/` (visited on 10/06/2026).

Drago, F., R. Galbiati, and P. Vertova (2009). "The Deterrent Effects of Prison: Evidence from a Natural Experiment". In: *Journal of Political Economy* 117.2, pp. 257–280.

Gelman, A. and G. W. Imbens (2019). "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs". In: *Journal of Business & Economic Statistics* 37.3, pp. 447–456.

Hahn, J., P. Todd, and W. van der Klaauw (2001). "Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design". In: *Econometrica* 69.1, pp. 201–209.

References III

Hansen, B. (2015). “Punishment and Deterrence: Evidence from Drunk Driving”. In: *American Economic Review* 105.4, pp. 1581–1617.

Hoekstra, M. (2009). “The Effect of Attending the Flagship State University on Earnings: A Discontinuity-Based Approach”. In: *Review of Economics and Statistics* 91.4, pp. 717–724.

Imbens, G. and K. Kalyanaraman (2012). “Optimal Bandwidth Choice for the Regression Discontinuity Estimator”. In: *Review of Economic Studies* 79.3, pp. 933–959.

Imbens, G. W. and J. D. Angrist (1994). “Identification and Estimation of Local Average Treatment Effects”. In: *Econometrica* 62.2, pp. 467–475.

Imbens, G. W. and T. Lemieux (2008). “Regression Discontinuity Designs: A Guide to Practice”. In: *Journal of Econometrics* 142.2, pp. 615–635.

Kolesar, M. and C. Rothe (2018). “Inference in Regression Discontinuity Designs with a Discrete Running Variable”. In: *American Economic Review* 108.8, pp. 2277–2304.

Lee, D. S. and D. Card (2008). “Regression Discontinuity Inference with Specification Error”. In: *Journal of Econometrics* 142.2, pp. 655–674.

Lee, D. S. and T. Lemieux (2010). “Regression Discontinuity Designs in Economics”. In: *Journal of Economic Literature* 48.2, pp. 281–355.

Lee, D. S. and J. McCrary (2017). “The Deterrence Effect of Prison: Dynamic Theory and Evidence”. In: *Advances in Econometrics* 38, pp. 73–146.

Lee, D. S., E. Moretti, and M. J. Butler (2004). “Do Voters Affect or Elect Policies? Evidence from the U.S. House”. In: *Quarterly Journal of Economics* 119.3, pp. 807–859.

References IV

McCrary, J. (2008). “Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test”. In: *Journal of Econometrics* 142.2, pp. 698–714.

Thistlethwaite, D. L. and D. T. Campbell (1960). “Regression-Discontinuity Analysis: An Alternative to the Ex Post Facto Experiment”. In: *Journal of Educational Psychology* 51.6, pp. 309–317.

van der Klaauw, W. (2002). “Estimating the Effect of Financial Aid Offers on College Enrollment: A Regression-Discontinuity Approach”. In: *International Economic Review* 43.4, pp. 1249–1287.